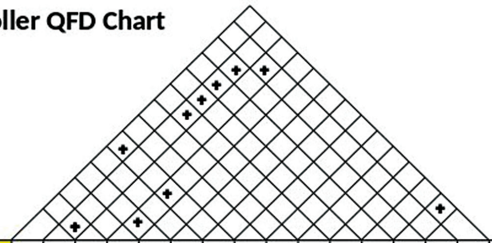
Charge Controller QFD Chart



Product Specification	Importance	Power Driver	CPU	Connectivity	Enclosure	Layout Plan	Data Storage	Warranty	Env Norm	Trade practice	Current Sensor	Volt sensor	Temp/Humid Sensor	Weather Sensors	Cloud storage	Analytics	Target
Operating Voltage	9	1															●
Max current	9	1															↑
Controller voltage	5		1														●
NEMA Std	7				1												↑
User interface	9			1													↑
Connectivity	7																↑
Size	5				1												↓
Aftersales Service	7					1											↑
Weight	5	1			1												↓
Cost	9		1	1			1	1									↓
Safety Margin	7							1									↑
Parts env. Norm	5								1								↑
Labor norm	3								1	1							↑
Sustainability Norm	4								1	1							↑
Current Sense	4		1								1						●
Voltage Sense	4		1									1					●
Temp sense	4		1										1				●
Humidity Sense	4		1										1	1			●
Data Storage Tech	5						1								1		●
Data Storage Size	5						1								1	1	↑
Importance		23	30	18	17	7	19	16	12	7	4	4	8	4	10	5	
Goal		●	●	●	●	●	↑ ↑	↑	↑ ●			●	●	↑ ↑	↑		

Fig. 4: Second QFD house—Design parameters

requirements. These features we put in the columns starting from the third one. We also put a mark '1' to indicate which product features address which requirements. There can be more than one feature to address a particular requirement. Also, one feature may address multiple requirements.

Next, we compare our target product with the ones that are available in the market. Competitions 1 to 4 are existing charge controllers. We took a weather station to compare the features of connectivity and user interface (Competition 5). We rank the product against the requirements. Shaded cells indicate the features that we selected for our product.

But, before selecting the target feature, we must complete the top portion, the roof. The triangular section shows how different product parameters are related. For instance, higher operating voltage has a negative effect on cost and safety margin. Higher current on the other hand will push up cost and safety margin.

Once we get a clear picture of the various interactions then we can select which required feature from the customer will be most suitable for our target product. As you see here, it is not necessary to take all the features from a single product. Some of the features we may decide to use can be our own.

Next, we compute the importance index of the parameters. These parameters address multiple important requirements. For instance, in our case, cost, followed by NEMA Standard, are the most important parameters. Based on the competition benchmarking we then decide our target values for each of the parameters.

Finally, we decide that our charge controller will be for a 48V alternator with a max current rating of 65A. It will have NEMA 3-level environment protection and housed in a 200mm × 180mm × 80mm box. It should weigh around 1kg and cost around ₹10,000. It will save data on an SD card and have a Wi-Fi connection to send operation data to a cloud server.

We must mention here that at this juncture the values that we select are just indicative figures that we picked up from the comparison of similar products from the market. Subsequent steps of design help us to validate, modify, and get further details of the product design.

In our next step, we use the product specification information to get the first-level design specifications. Product specifications are things that are visible to the customer. Design specifications are details that tell us how to make the product. Design specifications are the internal details that are not visible to the customer.

For getting the design specifications, we start with the list of product specifications. This is the set of features that needs to be delivered by our product. Next, we give an importance rating of the product specifications. Some practitioners recommend using the same importance figures that we got in the first chart.

Product design is primarily an engineering design exercise. One should not blindly follow the figures that have come from intuitions and group opinions. In my experience, re-evaluation and moderation of the importance figures in the second